

Imperial College London

BSc/MSci EXAMINATION Summer 2022

This paper is also taken for the relevant Examination for the Associateship

MRES - STATISTICAL METHODS FOR EXPERIMENTAL PHYSICS (MOCK EXAMINATION)

For MLBD MRes Students

June, 2022: 12:00 to 14:00

Answer all FOUR questions

Marks shown on this paper are indicative of those the Examiners anticipate assigning.

General Instructions

Complete the front cover of each of the 4 answer books provided.

If an electronic calculator is used, write its serial number at the top of the front cover of each answer book.

USE ONE ANSWER BOOK FOR EACH QUESTION.

Enter the number of each question attempted in the box on the front cover of its corresponding answer book.

Hand in 4 answer books even if they have not all been used.

You are reminded that Examiners attach great importance to legibility, accuracy and clarity of expression.

1. (i) A Bernoulli random variable X is a discrete random variable whose outcome is binary, often represented as a 0 or 1. The Bernoulli parameter p is the probability of ‘success’ or equivalently the probability that $X = 1$. We define it as $P(X = 1) = p$.
- (a) What is the expectation value of X when X is distributed as a Bernoulli random variable with parameter p ?
 - (b) What is the variance of X when X is distributed as a Bernoulli random variable with parameter p ?

[5 marks]

- (ii) The binomial distribution describes the probability to observe k successful outcomes in n independent trials,

$$P(k; n, p) = \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k}$$

where p is the probability that a single outcome is successful.

- (a) Show that the expectation value of the binomial distribution is np
- (b) Show that the variance of a binomial distribution is $np(1-p)$. You can use the fact that the variance of the sum of independent variables is the same as the sum of the variances of the independent variables.

[10 marks]

- (iii) We can estimate the binomial parameter p for a given observation of k and a fixed value n through both the method of moments and maximum likelihood estimator.

- (a) Show that the method of moments estimator for p is given by $\hat{p} = \frac{k}{n}$.
- (b) What is the maximum likelihood estimator for p ?
- (c) The “bias” of an estimator $\hat{\theta}$ is defined as the difference between the expectation value of the estimator and the value of the parameter being estimated θ ,

$$\text{bias}(\hat{\theta}) = E[\hat{\theta}] - \theta.$$

We say an estimator is “unbiased” if $\text{bias} = 0$. Show that the maximum likelihood estimator for the Bernoulli parameter p is unbiased.

- (d) Name any property apart from unbiased that makes an estimator a “good” estimator. Describe what that property means.

[10 marks]

[Total 25 marks]

2. (i) Hypothesis testing is the frequentist process of rejecting (or accepting) a scientific hypothesis given some observation in experimental data. Formally two hypotheses are defined, H_0 := the null hypothesis, and H_1 := the alternate hypothesis.

- (a) State the definition of Type-I and Type-II errors in terms of the two hypotheses H_0 and H_1 .
- (b) A testing apparatus is designed to detect for a faulty pieces of lab equipment. The test will report 'T' if the equipment is faulty and 'N' if the equipment is fine. The designer claims that the probability that the test reads 'T' when equipment is faulty is 80%. You test out a piece of equipment in your lab with the aparatus and get a reading 'T'. Which of the following statements is true given the information you have.
- The probability that the equipment is faulty is 80%.
 - If the equipment is not faulty, the probability that the test reports 'T' is 20%.
 - The probability that the test would have read 'N' if the equipment was faulty is 20%

[5 marks]

- (ii) An experiment is designed to detect decays of muon particles after being produced in an accelerator. A detector is setup to measure the decays time, t , after the the muon is produced $t_0 = 0$. The experiment runs until a maximum time T such that no decay times $t > T$ are recorded. The probability to observe a decay time t after is given by the exponential probability distribution,

$$f(t; \tau) = N \exp\left(-\frac{t}{\tau}\right)$$

where τ is the *lifetime* parameter.

- (a) Derive N as a function of τ
- (b) The figure below (figure 1) shows the distribution a distribution in data (observed values of the decay time t) collected by the experiment and an exponential probability distribution.
Describe how you would calculate the cumulative empirical distribution function $F_{emp}(t)$ for the data.
- (c) The Kolomogorov-Smirnov test-statistic is given by,

$$D = \max \left\{ |F_{emp}(t) - F(t)| \right\},$$

where $F(t)$ is the cumulative distribution of the exponential function in this case. Derive $F(t)$.

- (d) A test is performed to see if the exponential is a good description for the data shown. The figure below (figure 2) shows the distribution of D assuming the exponential shown is the true distribution ($f(D|H_0)$) and the observed value of the test statistic D_{obs} . The figure also shows a value D_{95} for which 95% of the distribution is to the left of the line. Describe how the information below could be used to test if the hypothesis that the exponential is a good description of the data is true or not.

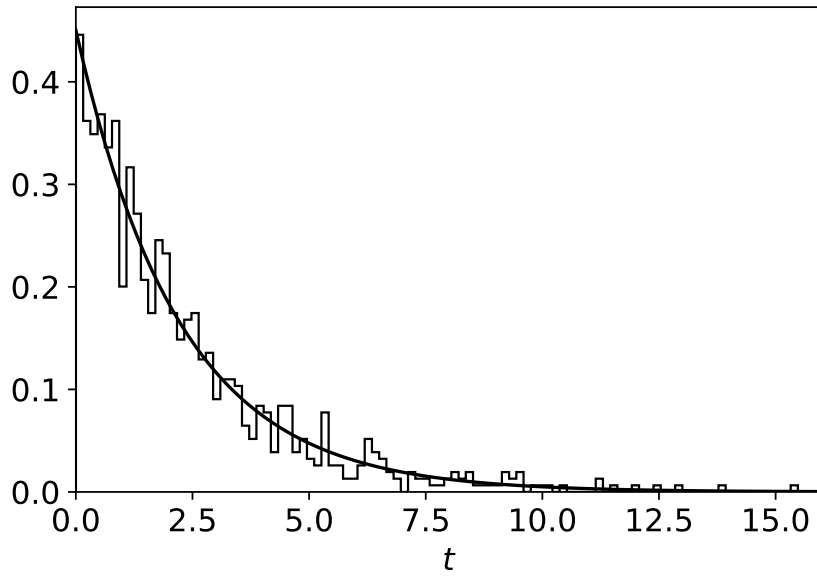


Figure 1

[10 marks]

- (iii) Suppose now we would like to measure the parameter τ from the observed data using the same experiment as in part (ii). Assume for the rest of this question that the times measured are much smaller than the maximum time T or equally that $T \rightarrow +\infty$.
- Write down the likelihood function for a set of n observations $\{t_1, t_2, \dots, t_n\}$ assuming the same exponential probability distribution from part (ii).
 - Derive the maximum likelihood estimate of the parameter $\hat{\tau}$
 - Explain how you would construct a Neyman interval at a 68% confidence level for the parameter τ . You should clearly state all of the steps required and you can write your answer in terms of pseudo-code if you wish.

[10 marks]

[Total 25 marks]

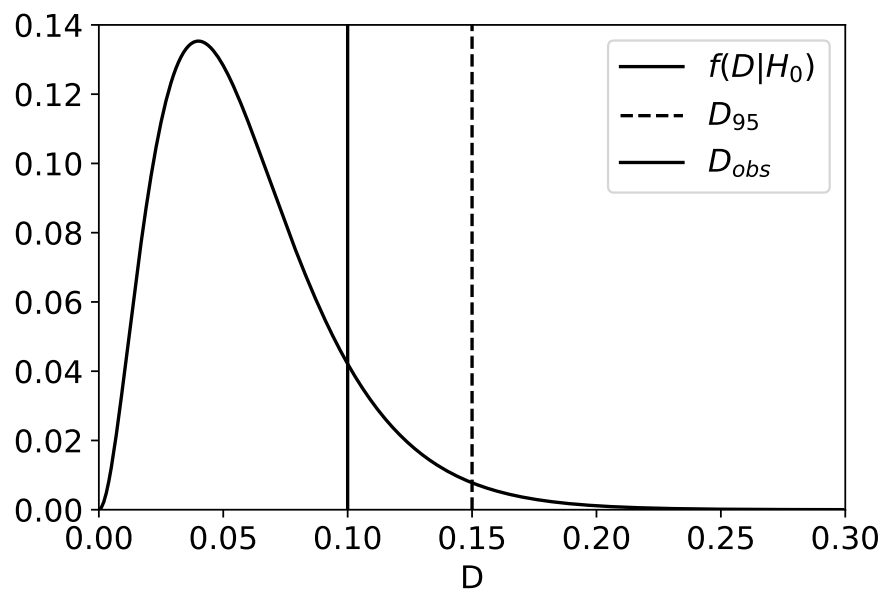


Figure 2

3. (i) (a) State Bayes theorem, naming all the terms when it is used for parameter inference of model parameters θ with some data d .
 (b) In what circumstances might you consider using a 'Jeffreys' prior $\pi(\theta) \propto 1/\theta$? [6 marks]

- (ii) An experiment is conducted to measure the number of photons coming from a reactor. The counts are Poisson distributed

$$p(n|\lambda) = \frac{\lambda^n e^{-\lambda}}{n!}$$

where λ is the expected number during the experiment.

- (a) Formulate this as a Bayesian inference problem to infer the rate λ given a measurement n . With a Jeffreys prior for λ find an expression for the posterior.
 (b) What is the most probable (MAP) value of λ ?
 (c) Write down an expression for $\bar{\lambda}$, the posterior mean of λ in terms of $p(\lambda|n)$, and show that

$$\bar{\lambda} = n.$$

You may use the result $\int_0^\infty \lambda^n e^{-\lambda} d\lambda = n!$

[8 marks]

- (iii) Find the variance of the posterior.

[4 marks]

- (iv) The Fisher matrix for parameters θ_α is

$$F_{\alpha\beta} \equiv \left\langle -\frac{\partial^2 \ln L}{\partial \theta_\alpha \partial \theta_\beta} \right\rangle,$$

where L is the likelihood. In this case the matrix is a single number, whose inverse gives the (minimum) expected value of the variance.

Show that the prediction for the variance from the Fisher analysis is $\sigma_\lambda^2 = n - 1$, and give two reasons for why it differs from the variance you calculated from the posterior.

[7 marks]

[Total 25 marks]

4. A measurement is made of a galaxy's recession speed v , to determine whether the Universe is expanding or not. Two models are considered:

- ssm: The Steady-State model: $v = 0$.
- bbm: The Big Bang model: $v = Hr$,

where r is the distance to the galaxy, which we assume that we know ($r = 1$ Mpc), and H is the Hubble constant, a parameter of the model. Throughout this question, use units of km s^{-1} for velocities, Mpc for distances, and $\text{km s}^{-1}\text{Mpc}^{-1}$ for H . Unlike normal practice, it is not necessary to quote the units for the solutions.

The recession speed of the galaxy is measured, giving $v = 90 \text{ km s}^{-1}$. It is known that the measurement error distribution is gaussian, with zero mean and standard deviation $\sigma = 90 \text{ km s}^{-1}$.

- (i) (a) The models are 'nested'. Explain what this means and why it applies in this case.
- (b) Assuming a uniform prior, what is the posterior of H , in the bbm? There is no need to normalise it at this stage,

[5 marks]

- (ii) (a) What is the most probable value of H in the bbm?
- (b) Calculate the posterior uncertainty on H .

[5 marks]

- (iii) Now we consider model selection, to see whether the data prefer the ssm or the bbm. For this we need to pay attention to the normalisation of the probability distributions.

- (a) Assume that the uniform prior on H extends from -200 to $+200 \text{ km s}^{-1}\text{Mpc}^{-1}$ and is zero outside this range. What is the (normalised) prior?
- (b) Write down the product of the likelihood and the prior, now properly normalised.

[4 marks]

- (iv) (a) Hence find an integral expression for the Bayesian Evidence for the bbm, and integrate it to show that

$$Z_{\text{bbm}}(v) \simeq \frac{1}{400},$$

where you should explain why it is approximate, and why the approximation will be accurate in this case.

- (b) Argue that in the ssm, the Bayesian Evidence is simply the likelihood.
- (c) Hence calculate the Bayes factor $B_{\text{bs}} \equiv Z_{\text{bbm}}/Z_{\text{ssm}}$ and say which model is favoured by the data.

[11 marks]

[Total 25 marks]